Kazi Nimat Hasan

Email : kazinimat3@gmail.com
Dhaka, Bangladesh
Cell : +88 01911-289207
Profile: nimat007.github.io/Nimat-portfolio/
linkedin.com/in/kazi-nimat-hasan-427852341

ACADEMIC SUMMARY

Biomedical Engineering graduate with interdisciplinary experience in bioinformatics, machine learning, medical image analysis, and biomaterials. Contributed to four research manuscripts and developed hands-on hardware projects, with leadership experience in student organizations and community initiatives.

EDUCATION

Khulna University of Engineering & Technology CGPA: 3.80/4.00
B. Sc. Biomedical Engineering Mar 2022- Aug 2026
Notre Dame College GPA: 5.00/5.00
Higher Secondary Certificate (HSC) December 2020

UNDERGRADUATE THESIS

Thesis Title: Integrated Transcriptomic Analysis for Host-Directed Target Discovery in Measles Virus Infection

- Integrated public bulk RNA-seq (GSE229685) and single-cell RNA-seq (GSE168775) GEO datasets to investigate host transcriptional responses to measles virus infection.
- Performed differential gene expression and cross-dataset analysis to characterize persistent responses and distinguish infected-cell from bystander-cell responses.
- Investigated protein–protein interaction networks using STRING and Cytoscape and conducted molecular docking using PyRx to support candidate host-target prioritization.

PUBLICATIONS

1. **K.N. Hasan**, S. A. Redhila, and G. M. M. Rahman, "Integrated Transcriptomic Analysis for Host-Directed Target Discovery in Measles Virus Infection," (manuscript in preparation)
2. M. A. Sayed, M. A. Kadir, N. Mostakim, M. J. Hossain, M. S. Alam, **K. N. Hasan**, and S. Hossain "Advanced image analysis for liver tumor detection and visualization in CT images using automated segmentation." (In preparation for resubmission)
3. M. A. Sayed, M. A. Kadir, M. S. Islam, A. Rahman, M. N. Hossain, J. Park, M. S. I. Wadud, R. T. Asha, **K. N. Hasan**, and H. Mehdi, "Enhancing lung cancer diagnosis with region-of-interest guided classification, segmentation, and PET-like feature extraction from CT scans," (In preparation for resubmission)
4. S. A. Redhila, S. Tasnim, **K. N. Hasan**, and K. Z. Ferdwushee, "Development and Optimization of Physically Cross-linked Chitosan/PVA/Gelatin Composite Hydrogels for Antibiotic-Loaded Wound Dressing Applications," (manuscript in preparation)

TECHNICAL EXPERTISE

Programming: Python, R, MATLAB, C

Bioinformatics & Computational Biology: Bulk & single-cell RNA-seq, Differential Expression (DESeq2, apeglm), scRNA-seq QC & Doublet Detection (Seurat, scDblFinder), Data Integration & Batch Correction (Harmony, RPCA), Pseudobulk Analysis, PPI Network Analysis (STRING, Cytoscape), Molecular Docking (PyRx)

Modeling & Simulation: COMSOL Multiphysics

Hardware & Prototyping: Arduino, ESP8266/NodeMCU, Sensor Integration, Embedded Systems, Bluetooth/RF Communication, IoT Prototyping

Biomaterials & Laboratory Methods: Chitosan/PVA/Gelatin Hydrogels, Freeze–Thaw Crosslinking, Freeze-Drying, Swelling, Gel Fraction & Degradation Assays

PROJECTS

Transcriptome-Based Drug Response Benchmark Sept 2026 – Present

- Benchmarked Ridge Regression and Random Forest on DepMap expression and PRISM 19Q4 drug-response AUCs (1,198 compounds) using five-fold lineage-grouped cross-validation to test generalization to unseen cancer lineages.
- Found that neither model consistently beat the mean-baseline MAE, quantifying the difficulty of cross-lineage prediction.

AI-Driven Glucose Monitoring for Telemedicine May 2026

- Developed a low-cost glucose estimator combining phototransistor-based sensing with a Random Forest model implemented in Python.

Pacemaker Electrode Design and Analysis (COMSOL Multiphysics) Jun 2024

- Simulated pacemaker electrode designs in COMSOL Multiphysics, comparing electrode materials and diameters to evaluate design performance.

Low-Cost Biomedical IoT & Assistive Devices 2023–2025

- Developed three Arduino/NodeMCU prototypes: a gesture-based assistive device for paralyzed patients using a custom self-made foil-graphite flex sensor, an IoT smart cradle monitoring crying and environment, and a patient health-monitoring system.

TRAINING & CERTIFICATIONS

Introduction to Genomic Technologies | Johns Hopkins University (Coursera) Oct 2026
Python for Genomic Data Science | Johns Hopkins University (Coursera) Oct 2026
Machine Learning with Python (80 hrs) | KUET Central Computer Center, BCC EDGE Project Oct–Dec 2024
Research Methodology: Basic to Advanced (48 hrs) | Research Help Bangladesh 2025
Basics of Bioinformatics Research: From Idea to Article | Udemy 2025

HONORS & AWARDS

Bronze Award for Voluntary Blood Donation | DREAM, KUET Nov 2025
BME Indoor 4.0: Champion, Table Tennis (Singles); Runner-up, Table Tennis (Doubles) Dec 2024
ALOHA Mental Arithmetic: International Grand Champion (Malaysia, 2013); 2× National Champion (2012, 2013)

EXTRACURRICULAR ACTIVITIES

Vice President, Biomedical Engineering Association, KUET Sep 2025 – Sep 2026

- Highest-ranking student representative in the department; organized the study tour, new-student orientation, a seminar on preventing sexual harassment, and BME Indoor 4.0.

Vice President, TRY (Voluntary Organization of KUETians) Aug 2025 – Aug 2026

- Led a winter clothing drive for 180 people and an Iftar program for 123 people across three madrasas, plus fundraising campaigns.

Head of Administration, KUET Adventure Club Mar 2023 – Jul 2026

- Organized the national event "Tales in Frames" (150+ participants); progressed from Executive to Assistant Secretary to Head of Administration.

Other roles: Junior Organizing Secretary, ABC KUET (Apr 2023 – Mar 2024); Class Representative, BME, KUET (Mar 2022 – Jan 2023)

REFERENCES

Dr. G.M. Mahmudur Rahman, Associate Professor, Dept. of Biomedical Engineering, KUET | mahmud@bme.kuet.ac.bd
Jeba Fairooz Rahman, Assistant Professor, Dept. of Biomedical Engineering, KUET | jeba@bme.kuet.ac.bd
Dr. Mostafa Zaman Chowdhury, Professor, Dept. of Electrical and Electronic Engineering, KUET | mzceee@yahoo.com